

40.016 The Analytics Edge

Censored Data Analysis

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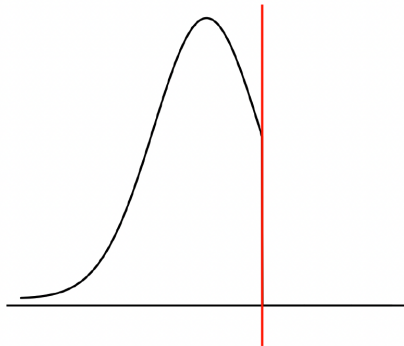
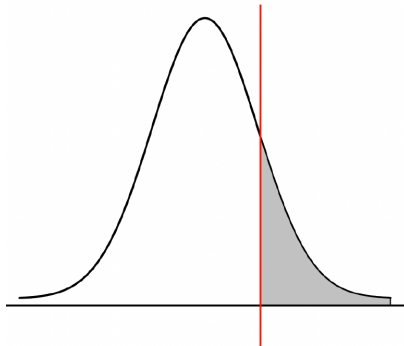
Censored Data

Where do we observe censored data?

- Survival of patients in a drug trial.
- Patience time of customers waiting in queue at call centers.
- Examinations/tests.
- Limitation of the measuring instrument.

Censored-Dependent Variable

- Suppose censored-dependent variable $Y^* \sim \mathcal{N}(\mu, \sigma^2)$.
- Consider censoring at C (say, capacity) above, so that we observe:



Agenda

- Social science: extra-marital affairs
 - Tobit model
- Healthcare: survival analysis
 - Kaplan-Meier estimation
 - Cox proportional hazards model

Tobit Model

Censored Tobit Regression

- Proposed by James Tobin (1958). Based on the regression model

$$\underbrace{Y_i^*}_{\substack{\text{Latent variable} \\ \text{(unobservable)}}} = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} + \epsilon_i, \forall i = 1, \dots, n,$$

- $\{\beta_1, \dots, \beta_p\}$: model coefficients
- $\{x_{i1}, \dots, x_{ip}\}$: predictor variables for the i -th individual
- n : the number of observations
- $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$: the model error.

The observable variable Y_i is left-censored at 0:

$$\underbrace{Y_i}_{\substack{\text{observable} \\ \text{variable}}} = \begin{cases} Y_i^*, & \text{if } Y_i^* \geq 0, \\ 0, & \text{if } Y_i^* < 0. \end{cases}$$

Censored Tobit Regression

- The density of Y given the vector of predictors $\mathbf{x} = (x_1, \dots, x_p)$ is the same as the density of Y^* given $\mathbf{x}$ for $Y^* > 0$.
- Further,

$$\begin{aligned}\mathbb{P}(Y = 0 \mid \mathbf{x}) &= \mathbb{P}(Y^* \leq 0 \mid \mathbf{x}) = \mathbb{P}(\epsilon \leq -\mathbf{x}^\top \boldsymbol{\beta} \mid \mathbf{x}) \\ &= \mathbb{P}\left(\frac{\epsilon}{\sigma} < -\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma} \mid \mathbf{x}\right) \\ &= \Phi\left(-\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right) = 1 - \Phi\left(\frac{\mathbf{x}^\top \boldsymbol{\beta}}{\sigma}\right),\end{aligned}$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)$, $\mathbf{x} = (1, x_1, \dots, x_p)$, and $\Phi(\cdot)$ is the standard normal cumulative distribution function.

Censored Tobit Regression

Let $(\mathbf{x}_i, y_i)$ be a random draw from the population. The density of Y given $\mathbf{x}_i$ is:

$$f_{Y|\mathbf{X}}(y_i | \mathbf{x}_i) = \frac{1}{\sigma} \phi \left(\frac{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right) \text{ for } y_i > 0,$$
$$\mathbb{P}(Y = 0 | \mathbf{x}_i) = \Phi \left(-\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right) = 1 - \Phi \left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right),$$

where ϕ is the standard normal density function.

Then the log-likelihood function for each observation i :

$$l_i(\boldsymbol{\beta}, \sigma) = \mathbf{1}(y_i = 0) \log \left[1 - \Phi \left(\frac{\mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right) \right] + \mathbf{1}(y_i > 0) \log \left[\frac{1}{\sigma} \phi \left(\frac{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\sigma} \right) \right].$$

Here, when $y_i > 0$, the exact value y_i is observed, so the likelihood contribution is the density $f_{Y|\mathbf{X}}(y_i | \mathbf{x}_i)$.

Censored Tobit Regression

- Then the log-likelihood function for each observation i :

$$l_i(\beta, \sigma) = \mathbf{1}(Y_i = 0) \log \left[1 - \Phi \left(\frac{\mathbf{x}_i^\top \beta}{\sigma} \right) \right] + \mathbf{1}(Y_i > 0) \log \left[\frac{1}{\sigma} \phi \left(\frac{y_i - \mathbf{x}_i^\top \beta}{\sigma} \right) \right].$$

- The log-likelihood for a random sample with n observations is obtained by summing $l_i(\beta, \sigma)$ across all $i \in \{1, \dots, n\}$.
- The maximum likelihood estimates of β and σ are obtained by maximizing the log-likelihood:

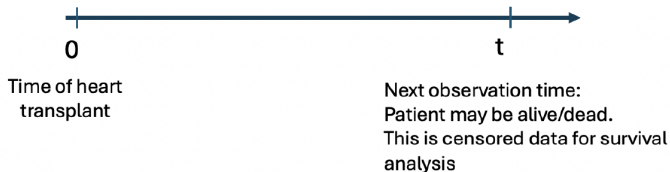
$$\max_{\beta, \sigma} \sum_{i=1}^n l_i(\beta, \sigma).$$

R Implementation

- `survival` package, with function for running the three models
- `survreg()`

Kaplan-Meier Estimator

Kaplan-Meier Estimator



- Suppose T = waiting time until a specific event happens (e.g., death).
- Let T have c.d.f $F(t)$ and p.d.f. $f(t)$.
- The *survival function* at time $t > 0$ is given by

$$S(t) = \mathbb{P}(T > t) = 1 - F(t).$$

Kaplan-Meier Estimator

- Proposed in 1958 by Kaplan and Meier
- A *non-parametric method* to estimate the survival function from censored lifetime data

Kaplan–Meier Estimator

- Let $t_1 \leq t_2 \leq \dots \leq t_N$ be the distinct observed event times
- d_i : number of events (e.g., deaths) at time t_i
- n_i : number of individuals **at risk** just before time t_i , i.e., individuals who have not experienced the event (are still alive)

The survival function can be estimated as

$$\hat{S}(t) = \prod_{i: t_i \leq t} \left(1 - \frac{d_i}{n_i}\right).$$

Kaplan–Meier Estimator

Heuristic Justification

- At time t_i :

$$\mathbb{P}(\text{death at } t_i \mid \text{alive before } t_i) \approx \frac{d_i}{n_i} \text{ and}$$

$$\mathbb{P}(\text{survive past } t_i \mid \text{alive before } t_i) \approx 1 - \frac{d_i}{n_i}.$$

- To survive to time t_i , one must survive:
 - Until t_1
 - From t_1 to t_2 , given survival up to t_1
 - And so on. Putting all together,

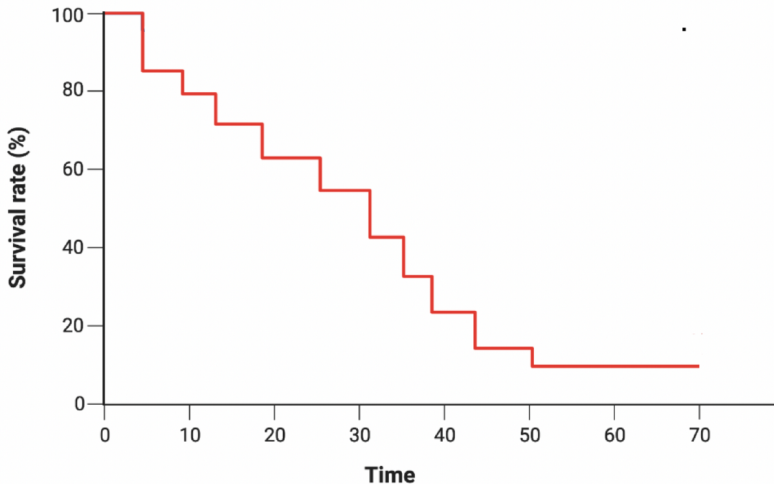
$$\hat{S}(t_i) = \left(1 - \frac{d_1}{n_1}\right) \cdot \left(1 - \frac{d_2}{n_2}\right) \cdot \dots \cdot \left(1 - \frac{d_k}{n_k}\right), \quad \text{for } t_k \leq t_i.$$

Kaplan–Meier Estimator

Example

- Consider the following remission times (in weeks) for 21 patients suffering from leukaemia:
- 1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23.

Kaplan–Meier Estimator



R Implementation

- `survfit()`

Cox Proportional Hazards Model

Cox Proportional Hazards Model

Hazard rate: Among individuals who have survived up to time t , how likely is the event to occur in the next instant? Formally, define hazard rate

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t < T \leq t + \Delta t \mid T > t)}{\Delta t}$$

Using conditional probability:

$$\lambda(t) = \frac{f(t)}{S(t)}$$

- T : time until the event occurs (a random variable)
- $f(t)$: probability density function of the event time T
- $S(t) = \mathbb{P}(T > t)$: survival probability

Cox Proportional Hazards Model

- Proposed by David Cox (1972).
- Used in bio-medical research (adopted in Framingham heart study)
- Estimates hazard rate and hence survival rate (at an individual level).
- Describe hazard rate $\lambda(t)$ as

$$\lambda(t) = \lambda_0(t) \left(e^{\beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}} \right) = \lambda_0(t) e^{\beta^\top \mathbf{x}_i}.$$

- $\{\beta_1, \dots, \beta_p\}$ = model coefficients,
- $\mathbf{x}_i = \{x_{i1}, \dots, x_{ip}\}$ = predictor variables for i -th individual.
- baseline hazard rate $\lambda_0(t)$: hazard where all predictors are 0 .

Cox Proportional Hazards Model

$$\lambda(t) = \lambda_0(t) \left(e^{\beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p} \right) = \lambda_0(t) e^{\beta^\top x}$$

- $\beta_j > 0$ means as $x_{ij} \uparrow$, we have $\lambda(t) \uparrow$. So length of survival $\downarrow$.

The likelihood of observing that individual i experiences the event at time t_i , among all individuals at risk at this time:

$$\ell_i(\beta) = \frac{\lambda_0(t_i) e^{\beta^\top x_i}}{\sum_{j \in R(t_i)} \lambda_0(t_i) e^{\beta^\top x_j}} = \frac{e^{\beta^\top x_i}}{\sum_{j \in R(t_i)} e^{\beta^\top x_j}},$$

- t_i : time at which individual i experiences the event
- $R(t_i)$: the set of all individuals at risk (who have not experienced the event) at time t_i

Then β can be estimated by solving an MLE problem.

R Implementation

- `coxph()`

Reference

- Teaching notes.